Greater Noida College

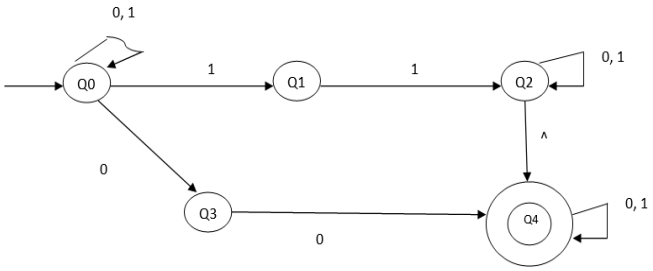
Assignment-1

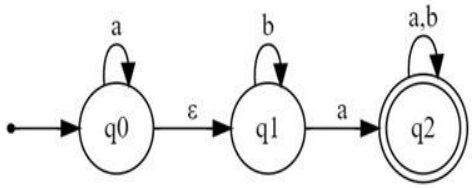
Subject- Theory of Automata & Formal Language

1. Give the mathematical definition of DFA
2. Construct DFA to accept string that always start with ‘ab’ over alphabet ∑ = {a,b}.
3. What do you mean by ∈- closure in NFA?
4. Define Alphabet and String in Automata Theory.
5. Explain in brief about the Kleen’s closure.
6. Construct DFA to accept string that always start with substring ‘aaa’ over alphabet ∑ = {a,b}.
7. Convert the following NFA $\{p,q,r,s\},\{0,1\},\delta,p,\{q,s\}$ into DFA where δ is given by

	0	1
→p	q, s	q
*q	r	q, r
R	s	p
*s	ϕ	p

8. What are various points of difference between Moore & Mealy Machine? Explain with example.
9. Design a FA to accept the string that always ends with 101.
10. How do we determine equivalence of two DFA? Explain with an example.
11. Construct a Non Deterministic Finite Automation (NFA) for the language L which accepts all the strings in which the third symbol from right end is always 'a' over Σ = {a, b}.
12. What do you mean by word Automata? Define types of Finite Automata.
13. Construct a DFA equivalent to the NFA



14. Construct a Finite automata (DFA) which accepts all binary numbers whose decimal equivalent is divisible by 3 over ∑= {a, b}.

15. Convert the following ∈- NFA to DFA